

RAG Evaluation

SaaS Intelligent AI Agent for Document Automation

IT Gate , Internship Project

I evaluated the retrieval and answer generation parts of the RAG pipeline using 14 test cases. The cases cover questions about my uploaded docs on the platform exp my cv..

Results

<i>Metric</i>	<i>Result</i>
<i>Evaluation cases</i>	<i>14</i>
<i>Evaluation Hit Rate</i>	<i>100%</i>
<i>Retrieval Recall</i>	<i>100%</i>
<i>Mean Reciprocal Rank (MRR)</i>	<i>88.1%</i>
<i>Answer Contains Score</i>	<i>83.3%</i>
<i>Token F1</i>	<i>67.4%</i>
<i>LLm-judged Answer correctness</i>	<i>96.4%</i>

Interpretation

1/Retrieval Hit Rate

Measures how often the system retrieves at least one relevant chunk for a question .

res = 100% means every one of 14 test cases had at least one relevant chunk retrieved

2/ Retrieval Recall

Measures how much of expected relevant info was retrieved.

res = 100% means the evaluation cases' expected relevant chunks were retrieved.

This is based on the ground-truth chunks defined in the evaluation cases.

3/ Mean Reciprocal Rank (MRR)

Measures how high first relevant result appears in retrieved list

Obtaining 88.1% means relevant info was not always ranked first

4/ Answer Contains Score

Measures whether required words or phrases from expected answers appear in generated ones.

83.3% → generated answers contained about 83% of required terms

5/ Token F1

compares words in generated answer with ones in the expected answers

Token F1= Combines precision (how much of the generated text overlaps) and recall (how much of the expected text is covered)

67.4% → Indicated moderate word-level overlap

Note : also this result may seem lower not because answer is wrong but → model may use different but correct wording

6/LLM-Judged Answer Correctness

i used the qwen model to judge and compare expected answers vs generated ones

res = 96.4 % → means most of cases were answered correctly

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